

SISTEMA DI RACCOMANDAZIONE



Prodotti cosmetici su similarità multidimensionale

*«La similarità deve riflettere
ingredienti, brand e categoria,
non solo il nome del prodotto.»*

CONTESTO

Il mercato cosmetico presenta un'elevata varietà di prodotti apparentemente simili ma con composizioni chimiche molto diverse.

I sistemi di raccomandazione tradizionali si basano spesso su descrizioni testuali o strategie di marketing, trascurando la reale formulazione del prodotto.

Questo approccio può portare a suggerimenti poco coerenti dal punto di vista funzionale e qualitativo.

OBIETTIVO

Sviluppare un sistema di raccomandazione in grado di identificare prodotti cosmetici realmente simili a un prodotto di partenza.

DATASET

Il dataset utilizzato proviene dalla piattaforma **OpenFoodFacts** ed è stato filtrato per includere esclusivamente prodotti cosmetici.

Il dataset originale è composto da circa **60.000 prodotti** e presenta dati grezzi, rumorosi e non normalizzati.

Prima dell'analisi è stata effettuata una fase approfondita di **pulizia e ristrutturazione** dei dati.

DATASET ROW

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
code	url	creator	created_t	created_d	last_modif	last_modif	last_modif	last_updat	last_updat	product_n	abbreviate	generic_n	quantity	packaging	packaging	packaging	packaging	brands	brands_tag	brands_en	categories	categories
00000001	http://world-en.openbeautyfacts.org/product/00000001	tonyracket1751609682	2025-07-04	T06:14:42Z	1751609686	2025-07-04	T06:14:46Z	tonyracket1751609686	2025-07-04	T06:14:46Z	Arnica	ait:arnica	ait:arnica	en:italy	en:italy	italy	en:to-be-completed,	en:				
00000014	http://world-en.openbeautyfacts.org/product/00000014	susslupinen-mehl-bodylab-24	insectproduct	add1710345659	2024-03-13	T16:00:59Z	1748277440	2025-05-26	T16:37:20Z	fitcounter1748277440	2025-05-26	T16:37:20Z	SÃ¼Ã¶lupinen Mehl	20pcs	Box							
00000018	http://world-en.openbeautyfacts.org/product/00000018	chocolate-peanut-butter-protein	foodvisor1729882037	2024-10-25	T18:47:17Z	1742914310	2025-03-25	T14:51:50Z	smoothie-app1742914310	2025-03-25	T14:51:50Z	Chocolate peanut butter protein										
00000092	http://world-en.openbeautyfacts.org/product/00000092	airpods-appleproduct-scan-com	1638580447	2021-12-04	T01:14:07Z	1744901851	2025-04-17	T14:57:31Z	smoothie-app1744901851	2025-04-17	T14:57:31Z	airpods1	Applexx:apple	apple	france	en:fr						
00000163	http://world-en.openbeautyfacts.org/product/00000163	smoothie-app1682257711	2023-04-23	T13:48:31Z	1727959822	2024-10-03	T12:50:22Z	fix-code-bot1743166885	2025-03-28	T13:01:25Z	Deutschland	en:germany	Germany	en:to-be-completed,	en:	nu						
00000280	http://world-en.openbeautyfacts.org/product/00000280	date-limite-app1584379762	2020-03-16	T17:29:22Z	1744884567	2025-04-17	T10:09:27Z	smoothie-app1744884567	2025-04-17	T10:09:27Z	Conserve	conserve	Conserve	en:France	en:france	France	en:					
00000311	http://world-en.openbeautyfacts.org/product/00000311	huile-vegetale-amande-douce-aroma-zone	openfoodfacts-contributors	1518297016	2018-02-10	T21:10:16Z	1727959827	2024-10-03	T12:50:27Z	fix-code-bot1743166886	2025-03-28	T13:01:26Z	Huile									
00000315	http://world-en.openbeautyfacts.org/product/00000315	colgate-sensitive-plus-made-in-india-oat-haus	foodvisor1656273028	2022-06-26	T19:50:28Z	1736518675	2025-01-10	T14:17:55Z	teolemon1743166980	2025-03-28	T13:03:00Z	Colgate Sensitive Plus										
00000320	http://world-en.openbeautyfacts.org/product/00000320	body-earth-incopenfoodfacts-contributors	1612887819	2021-02-09	T16:23:39Z	1747473474	2025-05-17	T09:17:54Z	teolemon1747473474	2025-05-17	T09:17:54Z	Body & Earth, Inc	Body & Earth	Inc	en:							
00000347	http://world-en.openbeautyfacts.org/product/00000347	quercetin-zinc-vitamin-c-vitamin-d-elderberry-natra	couticals-q-defend	foodvisor1714427298	2024-04-29	T21:48:18Z	1740528027	2025-02-26	T00:00:27Z	waistline-app1743166886	2025-03-28	T13:01:26Z										
00000423	http://world-en.openbeautyfacts.org/product/00000423	desmond1744616417	2025-04-14	T07:40:17Z	1748671427	2025-05-31	T06:03:47Z	smoothie-app1748671427	2025-05-31	T06:03:47Z	en:united-kingdom	en:united-kingdom	United Kingdom	en:to-be-completed,	en:							
00000493	http://world-en.openbeautyfacts.org/product/00000493	savon-vegetal-illuminateur-aroma-zone	openfoodfacts-contributors	1565650963	2019-08-12	T23:02:43Z	1727959829	2024-10-03	T12:50:29Z	fix-code-bot1743166886	2025-03-28	T13:01:26Z	savon									
00000511	http://world-en.openbeautyfacts.org/product/00000511	pure-collagene-marin-nullproduct-scan-com	1632331280	2021-09-22	T17:21:20Z	1727959748	2024-10-03	T12:49:08Z	fix-code-bot1743168706	2025-03-28	T13:31:46Z	pure collagÃ¨ne marin	120N									
00000561	http://world-en.openbeautyfacts.org/product/00000561	huile-vegetale-coco-aroma-zone	openfoodfacts-contributors	1518297122	2018-02-10	T21:12:02Z	1727959832	2024-10-03	T12:50:32Z	fix-code-bot1743166886	2025-03-28	T13:01:26Z	Huile vÃ©gÃ©									
00000563	http://world-en.openbeautyfacts.org/product/00000563	beaute-des-pieds-baume-reparateur-lavande-bio-yves-rocher	angela23751466793656	2016-06-24	T18:40:56Z	1727959832	2024-10-03	T12:50:32Z	fix-code-bot1743166886	2025-03-28	T13:01:26Z											
00000571	http://world-en.openbeautyfacts.org/product/00000571	openfoodfacts-contributors	17130	86550	2024-04-14	T09:22:30Z	1727959832	2024-10-03	T12:50:32Z	fix-code-bot1727959832	2024-10-03	T12:50:32Z	en:France	en:france	France	Olea europaea fruit						
00000587	http://world-en.openbeautyfacts.org/product/00000587	macerat-huileux-lys-aroma-zone	openfoodfacts-contributors	1518296822	2018-02-10	T21:07:02Z	1727959833	2024-10-03	T12:50:33Z	fix-code-bot1743166886	2025-03-28	T13:01:26Z	MacÃ©rÃ©									
00000592	http://world-en.openbeautyfacts.org/product/00000592	smoothie-app1744209612	2025-04-09	T14:40:12Z	1744209676	2025-04-09	T14:41:16Z	smoothie-app1744209676	2025-04-09	T14:41:16Z	en:south-africa	en:south-africa	South Africa	en:to-be-completed,	en:							
00000599	http://world-en.openbeautyfacts.org/product/00000599	smoothie-app1740311434	2025-02-23	T11:50:34Z	1740311587	2025-02-23	T11:53:07Z	smoothie-app1740311587	2025-02-23	T11:53:07Z	en:italy	en:italy	italy	en:to-be-completed,	en:	nutrition-facts-t						
00000659	http://world-en.openbeautyfacts.org/product/00000659	poires-chocolat-noir	kiliweb1558891084	2019-05-26	T17:18:04Z	1747809647	2025-05-21	T06:40:47Z	smoothie-app1747809647	2025-05-21	T06:40:47Z	Poires chocolat noir	en:FR	en:france	France	en:						
00000695	http://world-en.openbeautyfacts.org/product/00000695	ole-ole-aloe-loovara	openfoodfacts-contributors	1601906868	2020-10-05	T14:07:48Z	1727959748	2024-10-03	T12:49:08Z	fix-code-bot1743166885	2025-03-28	T13:01:25Z	LoÃ lÃ	oÃ lÃ	AloÃ	250mL	Lo					
00000735	http://world-en.openbeautyfacts.org/product/00000735	creme-pour-barbe-aliver	openfoodfacts-contributors	1542194248	2018-11-14	T11:17:28Z	1727959852	2024-10-03	T12:50:52Z	fix-code-bot1743166886	2025-03-28	T13:01:26Z	CrÃ©me pour barbe	30								
00000759	http://world-en.openbeautyfacts.org/product/00000759	gel-nail-polish-myleepaulin	em1685475941	2023-05-30	T19:45:41Z	1727959852	2024-10-03	T12:50:52Z	fix-code-bot1743166886	2025-03-28	T13:01:26Z	gel nail polish	10ml 0.33mylee	xx:mylee	mylee	en:						
00000834	http://world-en.openbeautyfacts.org/product/00000834	openfoodfacts-contributors	1532180415	2018-07-21	T13:40:15Z	1728653863	2024-10-11	T13:37:43Z	fix-code-bot1728653863	2024-10-11	T13:37:43Z	en:FR	en:france	France	en:to-be-completed,	en:						
00000859	http://world-en.openbeautvfacts.org/product/00000859	base-lavante-aroma-zone	openfoodfacts-contributors	1518296658	2018-02-10	T21:04:18Z	1727959834	2024-10-03	T12:50:34Z	fix-code-bot1743166886	2025-03-28	T13:01:26Z	Base lavante	Base	la							

FASE 1 Estrazione Brand

- Identificazione e selezione dei brand dal dataset originale
- Estrazione degli attributi chiave (codice, nome, brand, segmento)
- Rimozione di duplicati e record inconsistenti
- Standardizzazione dei formati dati

FASE 2 Estrazione Ingredienti

- Analisi della composizione per ogni prodotto
- Estrazione degli ingredienti (INCI names)
- Aggregazione degli ingredienti per codice prodotto
- Creazione di rappresentazioni testuali unificate

FASE 3 Categorizzazione

- Inserimento della macro-categoria per ogni tabella
- Mappatura prodotti-categorie (Deodorants, Hair Care, Face Care, ecc.)
- Verifica della coerenza categoriale
- Creazione di dataset finale integrato



Sistema di Raccomandazione One-Time



Approccio innovativo che evita la costruzione di matrici NxN, calcolando la similarità on-demand.



Filtro hard sulla macro-categoria



Combinazione pesata: 0.8 ingredienti + 0.2 brand/segmento



Array 1xK per prodotto query

SIMILARITA' INGREDIENTI



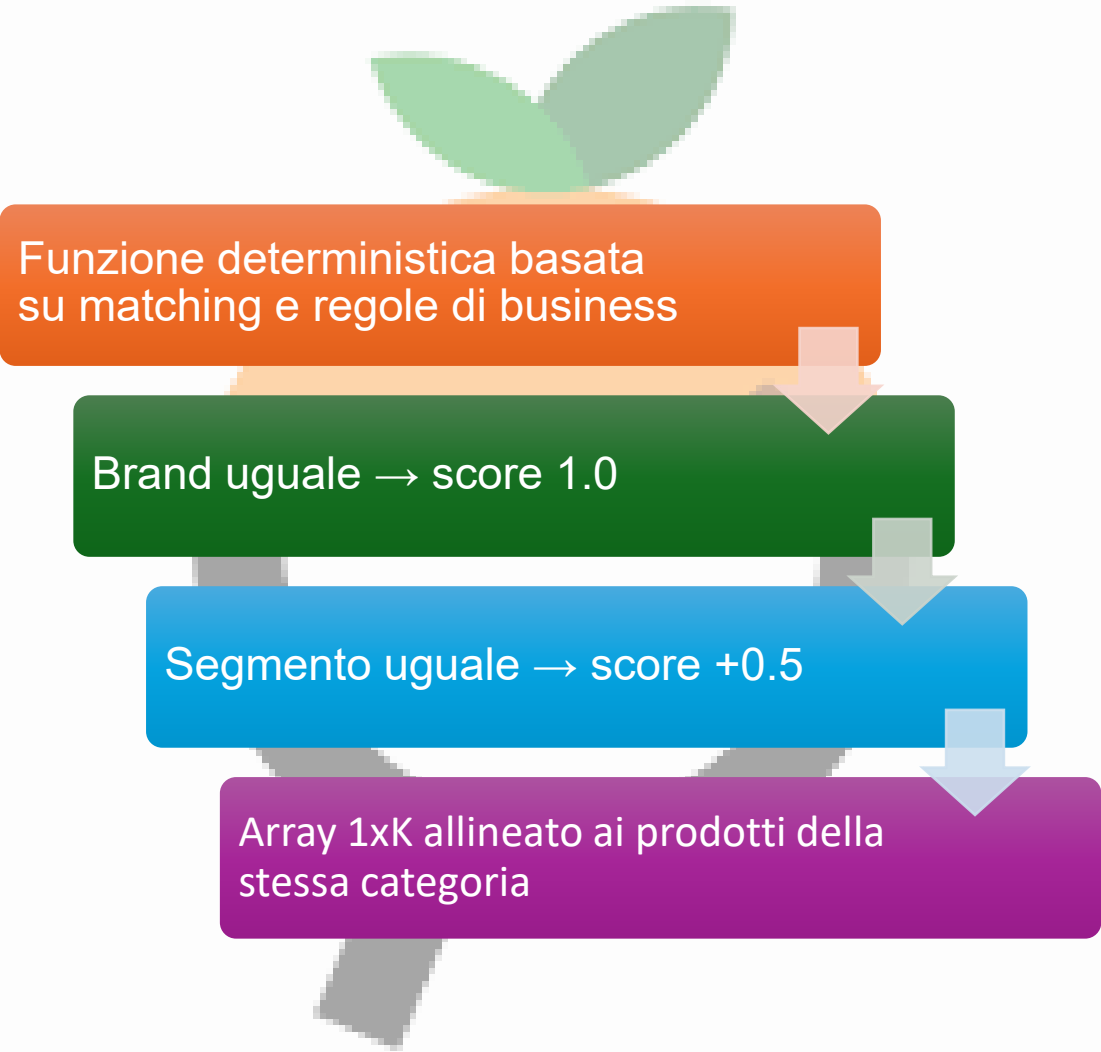
CountVectorizer su ingredienti
aggregati per prodotto

Rappresentazione bag-of-
words (prodotti $\times$ ingredienti)

Similarità coseno tra vettore
query e candidati

Array 1xK allineato ai prodotti
della stessa categoria

SIMILARITA' BRAND



Funzione deterministica basata
su matching e regole di business

Brand uguale → score 1.0

Segmento uguale → score +0.5

Array 1xK allineato ai prodotti della
stessa categoria

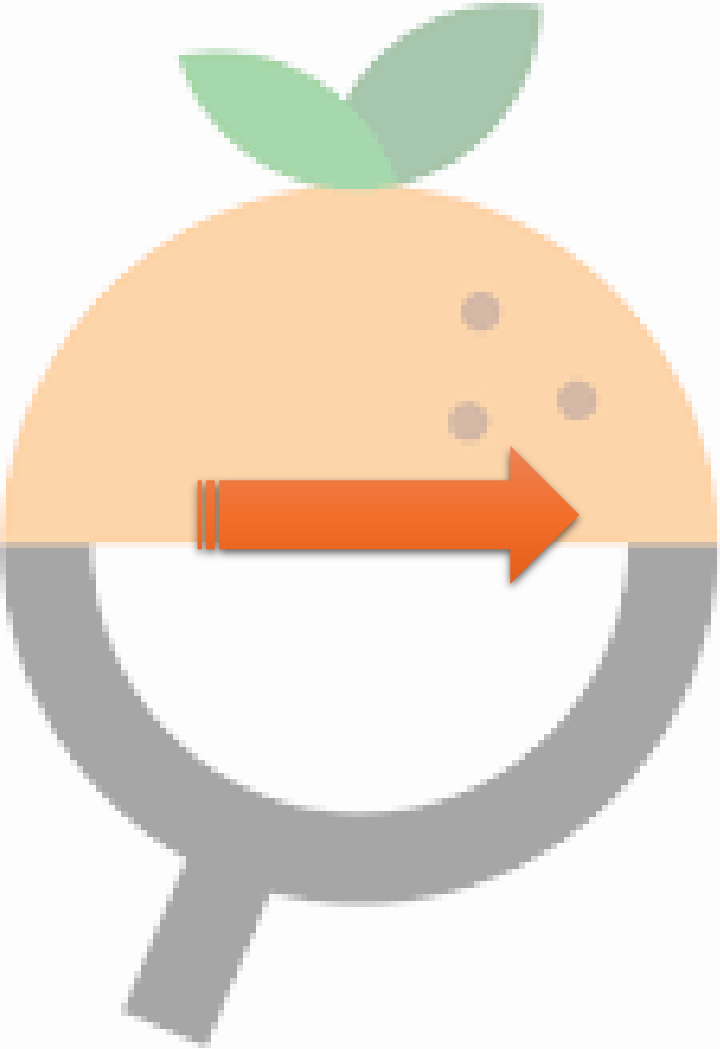
COMBINAZIONE

SCORE FINALE

0.8
x
sim_ingredienti

+

0.2
x
sim_brand



Gli array sono **perfettamente allineati** per prodotto, permettendo combinazione lineare diretta.

OUTPUT



Top-K raccomandazioni ordinate per score finale

Visualizzazione **score finale** ingredienti e brand

Lista **ingredienti comuni** tra query e candidate

ESEMPIO OUTPUT 1

rank	query_product_id	query_product_name	recommended_product_id	product_name	brand_name	brand_segment	common_ingredients	final_similarity_score
1	0718334312125	soap bar exfoliating italian lemon	0708002100543	tea tree soap bar	faith in nature	middle	[sodium gluconate, glycerin, sodium palm kerne...	0.540000
2	0718334312125	soap bar exfoliating italian lemon	3700376300434	savon extra doux	savonnerie aubagnaise	mass_market	[sodium gluconate, glycerin, sodium palm kerne...	0.500000
3	0718334312125	soap bar exfoliating italian lemon	3245678671139	savon doux a l huile d olive	cosmia	middle	[sodium palmate, sodium palm kernelate, oil, g...	0.495556
4	0718334312125	soap bar exfoliating italian lemon	3504750002019	savon au monoi tiare	tiki tahiti	middle	[sodium palmate, sodium palm kernelate, glycer...	0.495556
5	0718334312125	soap bar exfoliating italian lemon	3286011149144	savon huile d argan bio	biopha	middle	[sodium palmate, sodium palm kernelate, oil, g...	0.495556
6	0718334312125	soap bar exfoliating italian lemon	3284410017262	savon extra riche	melvita	middle	[sodium palmate, sodium palm kernelate, oil, g...	0.482857
7	0718334312125	soap bar exfoliating italian lemon	8000137010929	aloe detersione naturale sapone 100 vegetale	equilibra	middle	[sodium palmate, sodium palm kernelate, oil, g...	0.460000
8	0718334312125	soap bar exfoliating italian lemon	8025796001538	saponetta vegetale allo zolfo	i provenzali	middle	[sodium palmate, sodium palm	0.460000

ESEMPIO OUTPUT 2

rank	query_product_id	query_product_name	recommended_product_id	product_name	brand_name	brand_segment	common_ingredients	final_similarity_score
1	8001841617602	repair care	8001090094148	pantene pro v repair care	pantene pro v	mass_market	[trihydroxystearin, methylisothiazolinone, tet...	0.672727
2	8001841617602	repair care	8001090093936	pantene pro v color protect	pantene	mass_market	[trihydroxystearin, methylisothiazolinone, tet...	0.654286
3	8001841617602	repair care	8700216421102	rigenera e protegge	panthene	mass_market	[trihydroxystearin, tetrasodium edta, stearyl ...	0.557391
4	8001841617602	repair care	5410076651627	champoings	pantene	mass_market	[trihydroxystearin, tetrasodium edta, stearyl ...	0.509231
5	8001841617602	repair care	8001090662231	coconut milk	herbal essences	mass_market	[methylisothiazolinone, tetrasodium edta, sodi...	0.495556
6	8001841617602	repair care	4015600869588	swagger	old spice	mass_market	[trihydroxystearin, methylisothiazolinone, tet...	0.482857
7	8001841617602	repair care	7500435114912	pantene prov	pantene	mass_market	[tetrasodium edta, sodium benzoate, methylchloro...	0.453043
8	8001841617602	repair care	8001090047151	head shoulders 2in1 classic clean	head	mass_market	[sodium benzoate, methylchloroisothiazolinone,...	0.447692
9	8001841617602	repair care	8001090098054	classic	head	mass_market	[tetrasodium edta, sodium benzoate, methylchloro...	0.447692
10	8001841617602	repair care	4015600869236	danger time	old spice	mass_market	[methylisothiazolinone, sodium benzoate, sodiu...	0.440000

ESEMPIO OUTPUT 3

rank	query_product_id	query_product_name	recommended_product_id	product_name	brand_name	brand_segment	common_ingredients	final_similarity_score
1	8809153100634	black sugar mask wash off	3466762106225	masque confort apaisant fleur de coton reglisse	nocibe	mass_market	[euphorbia cerifera candelilla wax, butyrosper...	0.250345
2	8809153100634	black sugar mask wash off	7640183490279	hydration recovery masks set of 5 masks	swiss toniq geneva	mass_market	[capryliccapric triglyceride, oil, glycerin]	0.234118
3	8809153100634	black sugar mask wash off	3600542097024	skin active masque tissu charbon vegetal	l'oreal	mass_market	[capryliccapric triglyceride, oil, glycerin, bht]	0.220000
4	8809153100634	black sugar mask wash off	3574661575209	masque hydratation legerete 4 en 1	le petit marseillais	mass_market	[oil, butylparaben, glycerin, bht]	0.220000
5	8809153100634	black sugar mask wash off	3574661479699	masque 3 en 1 hydratation	le petit marseillais	mass_market	[oil, glycerin]	0.212727
6	8809153100634	black sugar mask wash off	4011609847466	q10 anti falten maske	maxim markenprodukte gmbh	mass_market	[oil, glycerin]	0.212727
7	8809153100634	black sugar mask wash off	8691988010383	sac maskesi	benri	mass_market	[oil, glycerin, bht]	0.209565



GRAZIE PER L'ATTENZIONE

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